

CHEMISTRY EXAM-02 — DETAILED MARKING REPORT

Chapters 3, 4, 5 — Atomic Structure, Periodic Table, Chemical Bonding

Seemab · Class 9 · Attempt Date: 2026-04-17

Final Score: 59.5 / 65 · 91.5% · Grade A+

12 / 12

Section A (MCQs)
100%

30.5 / 33

Section B (Short Qs)
92.4%

17 / 20

Section C (Long Qs)
85%

SECTION A — Multiple Choice Questions (12 / 12)

#	Question (summary)	Correct	Seemab's	Result
1	Subatomic particle with no charge	C (Neutron)	C	1
2	Atomic number = number of?	B (Protons)	B	1
3	Neutrons in $^{37}_{17}\text{Cl}$	C (20)	C	1
4	Max electrons in p sub-shell	B (6)	B	1
5	Group 1 elements are called	C (Alkali metals)	C	1
6	Number of periods in modern periodic table	C (7)	C	1
7	Highest electronegativity element	B (Fluorine)	B	1
8	Atomic size across a period	B (Decreases)	B	1
9	Bond formed by complete electron transfer	B (Ionic)	B	1
10	Shared pairs in N_2	C (Three)	C	1
11	Compound with coordinate bond	C (NH_4^+)	C	1
12	Hydrogen bonding with	B (N, O, F)	B	1

Perfect score on MCQs. Seemab demonstrated solid understanding of every conceptual touchpoint across Atomic Structure, Periodic Trends, and Chemical Bonding.

SECTION B — Short Questions (30.5 / 33)

Q.2(i) — Dalton's atomic theory & its importance 3 / 3

Seemab wrote: All 4 postulates correctly (atoms indivisible, atoms of same element identical, combine/rearrange in simple ratios, neither created nor destroyed). Importance clearly stated — foundation of modern chemistry, atom as building block.

Excellent answer. Structured and complete.

Q.2(ii) OR — Atomic number vs mass number 3 / 3

Seemab wrote: Clean comparison table — definitions, symbols (Z and A), example with $^{23}_{11}\text{Na}$ (Z=11, A=23).

Solid. Minor extra-working arithmetic slip (wrote " $23 - 11 = 20$ " instead of 12) but this wasn't part of the required answer. Full marks.

Q.2(iii) — Isotopes + three isotopes of hydrogen 3 / 3

Seemab wrote: Correct definition. Tabulated all three hydrogen isotopes (protium ^1_1H , deuterium ^2_1H , tritium ^3_1H) with protons 1/1/1, electrons 1/1/1, neutrons 0/1/2.

Exemplary. Clear tabular layout.

Q.2(iv) OR — Electronic configurations of $_{11}\text{Na}$, $_{13}\text{Al}$, $_{17}\text{Cl}$ 2 / 3

Seemab wrote: $\text{Na} = 1s^2, 2s^2, 2p^6, 3s^1$ ✓. Then labeled two answers "Al" — one with Z=17 config ($1s^2, 2s^2, 2p^6, 3s^2, 3p^5$ — which is Cl) and another with Z=13 config ($1s^2, 2s^2, 2p^6, 3s^2, 3p^1$ — which is actual Al).

Label mix-up: Seemab got all three configurations numerically right, but labeled both Cl's and Al's result as "Al". Lost 1 mark for mislabeling. Lesson: always double-check that the element symbol matches the Z value you used.

Q.2(v) OR — Group vs Period 3 / 3

Seemab wrote: Definitions (vertical rows = groups, horizontal = periods), quantities (18 groups, 7 periods), group → valence electrons, period → properties change gradually.

Complete and well-organized.

Q.2(vi) OR — Ionization energy definition + trends 2 / 3

Seemab wrote: Correct definition. Correct formula. Across period: "increases" ✓. Down group: heading written but **answer left blank** — no direction or reason stated.

Incomplete: The "down a group" trend (decreases, because atomic size and shielding increase) was started but never finished. Lost 1 mark. Always finish every sub-part before moving on.

Q.2(vii) OR — Why noble gases are inert 3 / 3

Seemab wrote: Group 18, complete valence shell ($ns^2 np^6$ except He $1s^2$), octet complete, don't participate in ordinary reactions, examples He, Ne, Ar configurations listed.

Thorough. Showed configurations with all 3 examples.

Q.2(viii) — Metals vs non-metals (3 differences) 3 / 3

Seemab wrote: Thermal conductivity (high vs poor), electrical conductivity (good due to free electrons vs poor), malleability/ductility, melting/boiling points (high vs low). 4 differences given.

Exceeds requirement. Free-electron reasoning included.

Q.2(ix) — Octet & Duplet rules with examples 3 / 3

Seemab wrote: Octet rule defined. Example: Na → Ne config, Cl → Ar config. Duplet rule defined. Example: Li → He config ($1s^2$).

Clear, includes configuration arrows showing electron transfer.

Q.2(x) OR — Polar vs Non-polar covalent bonds 2.5 / 3

Seemab wrote: Definitions (different vs same atoms, unequal vs equal sharing). Charge distribution (δ^+/δ^- vs symmetrical). Examples: H–H, O=O, **Cl=Cl**, N≡N (non-polar); Cl–H, H–O–H, HCN, NH₃ (polar).

Small error: Cl₂ has a single bond, should be **Cl–Cl**, not Cl=Cl. Everything else is accurate. Lost 0.5 mark.

Q.2(xi) OR — Hydrogen bonding in water 3 / 3

Seemab wrote: Clear definition (interaction between electron-deficient H and lone pair on electronegative atom like N, O, F). Water: O pulls electrons (δ^-), H becomes δ^+ , H of one molecule attracted to lone pair of O on another. Mentioned consequences — high boiling point, surface tension, universal solvent.

Excellent — went beyond required depth to link to observable properties of water.

SECTION C — Long Questions (17 / 20)

Q.3(i) OR — Rutherford's gold foil experiment + conclusions + defects 4 / 5

Seemab wrote: Experiment described (gold foil 0.00004 cm, α -particles from polonium = He²⁺, most slightly deflected, ~1 in million deflected >90°). **Conclusions:** most space empty, few deflected due to electrostatic repulsion from positive nucleus, massive α not deflected by electrons. **Defects:** (1) classical physics predicts electron should emit energy continuously and spiral into nucleus; (2) continuous spectrum expected but hydrogen shows line spectrum.

Missing: 3rd defect — Rutherford's model couldn't explain atomic stability or electron arrangement for multi-electron atoms. Lost 1 mark. The two defects she did give were excellently explained.

Q.3(ii) OR — Isotopes of H, C, Cl with abundance and uses 4 / 5

Seemab wrote: Defined isotopes. Tabulated all three elements:

- **Carbon:** C-12 (98.9%), C-13 (1.1%), C-14 (0.0009%) with p/e/n counts
- **Hydrogen:** protium (0 n, 99.99%), deuterium (1 n, 0.0015%), tritium (2 n, Rare)
- **Chlorine:** Cl-35 (18 n, 75.77%), Cl-37 (20 n, 24.23%)

Missing: The question also asks for **uses** of these isotopes (e.g., C-14 for carbon dating, deuterium in heavy water, Cl isotopes for chemical mass). Lost 1 mark for not stating uses. Great tabular organization otherwise.

Q.3(iii) — Periodic trends: atomic size, IE, EA, EN 4 / 5

Seemab wrote:

- **Atomic size:** Across period → decreases ✓ (nuclear charge ↑). **Down group → "decreases"** ← wrong direction. (Explanation "new electron shells added" is correct reasoning.)
- **Ionization energy:** Across → increases ✓, Down → decreases ✓
- **Electron affinity:** Across → increases ✓, Down → decreases ✓
- **Electronegativity:** F highest (4.0), Cs lowest (0.7) ✓

Important error: Atomic size **INCREASES** down a group (new shells added, electrons farther from nucleus). Seemab wrote "decreases" — contradicts her own correct reasoning. Lost 1 mark. All other trends are perfectly correct.

Q.3(iv) — Covalent bond formation: H₂, O₂, N₂ 5 / 5

Seemab wrote: Defined covalent bond (mutual electron sharing between non-metals). Explained single/double/triple bond notation.

- H₂: two H share one pair → H–H (to attain He duplet) ✓
- O₂: each O has 6 valence e⁻, need 2 more; share 2 pairs → O=O ✓
- N₂: each N has 5 valence e⁻, need 3 more; share 3 pairs → N≡N ✓

Perfect. All three bond types explained with clear electron-dot-and-cross representation.

Strengths

- **MCQs — 100% clean**, no guesswork evident
- Dalton's theory, hydrogen isotopes, hydrogen bonding, covalent bond formation — all A+ territory
- Tabular organization is consistent and easy to follow
- Goes beyond requirement (Q.2(viii) gave 4 differences, Q.2(xi) mentioned water's unique properties)
- Octet/duplet rule and its application shown with correct configuration mapping

Areas to Revise

- **Atomic size down a group** — memorize "shells add → size increases"; easy to mix up with "nuclear charge increases"
- **Ionization energy trend down a group** — finish the sentence! Don't leave sub-parts blank
- **Chlorine bond type** — Cl₂ is **single bond** (Cl–Cl), not double
- **Element labels** — double-check that the symbol in front matches the Z you're configuring (Al=13, Cl=17)
- **"Uses" of isotopes** — remember: C-14 (dating), deuterium (heavy water/reactors), Cl (trace analysis), Co-60 (cancer therapy), I-131 (thyroid)
- **Rutherford's 3rd defect** — couldn't explain multi-electron atoms or atomic stability

Overall: Seemab demonstrated strong mastery of Ch 3–5 core concepts. The errors were small and recoverable — mostly careless slips (atomic size direction, Cl=Cl, incomplete sentence) rather than conceptual gaps. With five minutes of revision on each flagged item, she would score 64–65/65.